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Fraction $\Rightarrow$ A part of a whole
{पूर्ण का एक हिस्सा}

like, $\frac{2}{6} \begin{matrix} \rightarrow \text{part} \\ \rightarrow \text{whole} \end{matrix}$

$\frac{x}{y}$ $x \rightarrow$ upper part is called Numerator
$y \rightarrow$ lower part is called Denominator

Q1. $\frac{132}{198} = ?$

A) 2/3 B) 3/2 C) 4/5 D) 11/15 E) 5/6

$\frac{132}{198}$ $132 \rightarrow$ Numerator, $198 \rightarrow$ Denominator

$= \frac{132}{198} = \frac{\cancel{11} \times 4 \times \cancel{3}}{\cancel{11} \times 6 \times \cancel{3}} = \frac{4}{6} = \frac{2}{3}$

Q2. $-\frac{150}{210} = ?$

A) −7/5 B) 5/7 C) −5/7 D) −3/7 E) −2/3

$\frac{-15\cancel{0}}{21\cancel{0}} = -\left(\frac{5 \times \cancel{3}}{7 \times \cancel{3}}\right) = -\frac{5}{7}$

Q3. $\frac{5}{12} = \frac{?}{96}$

A) 32 B) 36 C) 44 D) 40 E) 30

$\frac{5}{12} = \frac{x}{96}$

$\frac{5 \times \cancel{96}^{8}}{\cancel{12}} = x$

$x = 40$

Q4. $\frac{?}{45} = \frac{7}{15}$

A) 14 B) 18 C) 27 D) 24 E) 21

$\frac{x}{45} = \frac{7}{15}$

$x = \frac{7}{\cancel{15}} \times \cancel{45}^{3}$

$x = 21$

Q5. $\frac{9}{?} = \frac{3}{14}$

A) 28 B) 42 C) 36 D) 48 E) 54

$$\frac{9}{x} = \frac{3}{14} \qquad x = \frac{9}{3} \times 14$$

$$x = 3 \times 14$$

$$x = 42$$

Q6. If $x \times \left(\frac{8}{9}\right) = \frac{16}{27}$, then x = ?

A) 1/3 B) 2/3 C) 3/2 D) 9/8 E) 8/9

$$x \times \frac{8}{9} = \frac{16}{27}$$

$$x = \frac{16}{27} \times \frac{9}{8} = \frac{16}{8} \times \frac{9}{27} = 2 \times \frac{1}{3} = \frac{2}{3}$$

Q7. If $\frac{a}{b} = \frac{4}{9}$ and a = 28, then b = ?

A) 56 B) 84 C) 72 D) 63 E) 81

$$\frac{a}{b} = \frac{4}{9}, \quad \frac{28}{b} = \frac{4}{9}$$

$$\{a = 28\} \qquad b = \frac{28}{4} \times 9$$

$$b = 7 \times 9$$

$$b = 63$$

Mixed ↔ Improper

$\frac{\text{Numerator}}{\text{Denominator}}$.

Q8. $4\frac{7}{9} = ?$

A) 41/9 B) 42/9 C) 43/9 D) 44/9 E) 45/9

$$4\frac{7}{9} = 4 + \frac{7}{9} = \frac{36+7}{9} = \frac{43}{9}$$

$\boxed{\text{OR}}$

$$4\frac{7}{9} = \frac{(4 \times 9) + 7}{9} = \frac{36+7}{9} = \frac{43}{9}$$

when
Numerator < Denominator
It's Proper fraction.

like, $\frac{2}{3}, \frac{3}{7}, \frac{9}{10}, \frac{11}{25}$ etc.

when
Numerator > Denominator.
It's Improper fraction

like, $\frac{37}{7}, \frac{11}{5}, \frac{61}{2}$ etc.

Mixed fraction $\Rightarrow$ Fraction + Integer.

$\left\{ \text{Improper Fraction} > 1 \right\}$

Q9. $6\frac{5}{12} = ?$

A) $\frac{77}{12}$ ✓ B) $\frac{75}{12}$ C) $\frac{79}{12}$ D) $\frac{72}{12}$ E) $\frac{71}{12}$

$$6\frac{5}{12} = 6 + \frac{5}{12}$$

$$\frac{(12 \times 6) + 5}{12} = \frac{72 + 5}{12}$$

$$= \frac{77}{12}$$

$\boxed{\text{OR}}$

$$6\frac{5}{12} = \frac{(6 \times 12) + 5}{12}$$

$$= \frac{72 + 5}{12} = \frac{77}{12}$$

Q10. $\frac{58}{7} = ?$

A) $7\frac{2}{7}$ B) $8\frac{1}{7}$ C) $8\frac{2}{7}$ ✓ D) $8\frac{3}{7}$ E) $9\frac{2}{7}$

$\frac{58}{7} \rightarrow$ Improper Fraction

8 → Quotient

Divisor ← 7) 58 (

56

2 → Remainder

$$= 8\frac{2}{7} \left\{ \text{Mixed Fraction} \right\}$$

Q11. $-\frac{73}{8} = ?$

A) $-10\frac{1}{8}$ B) $-8\frac{1}{8}$ C) $-9\frac{5}{8}$ D) $9\frac{1}{8}$ E) $-9\frac{1}{8}$ ✓

$-\frac{73}{8}$ $\left\{ \text{Improper Fraction} \right\}$

9

8) 73 (

72

1

$$= -\left(9\frac{1}{8}\right)$$ Mixed Fraction

Q12. Find the greater fraction

$\left(\frac{7}{12}, \frac{5}{9}\right)$ $\frac{7}{12}, \frac{5}{9}$

Method 1 :- Use Cross Multiplication

$\frac{7}{12} \times \frac{5}{9}$

$63 > 60$

So, $\frac{7}{12} > \frac{5}{9}$

OR

Method 2 :-

$\frac{7}{12}, \frac{5}{9}$

Denominators को equal करना है !

$\frac{7 \times 3}{12 \times 3}, \frac{5 \times 4}{9 \times 4}$

$\frac{21}{36} > \frac{20}{36}$

So, $\frac{7}{12} > \frac{5}{9}$.

Q13. Find the smaller fraction

$\left(\frac{11}{15}, \frac{3}{4}\right)$

$\frac{11}{15} \times \frac{3}{4}$

$(11 \times 4) \quad (3 \times 15)$

$44 < 45$

OR

$\frac{11 \times 4}{15 \times 4}, \frac{3 \times 15}{4 \times 15}$

$\frac{44}{60} < \frac{45}{60}$

Q. Find the larger fraction

$\left(\frac{22}{7}, \frac{28}{9}\right)$

$\frac{22}{7} \times \frac{28}{9}$

$(22 \times 9) \quad (28 \times 7)$

$198 > 196$

OR

$\frac{22 \times 9}{7 \times 9} ; \frac{28 \times 7}{9 \times 7}$

$\frac{198}{63} > \frac{196}{63}$

So, $\frac{22}{7} > \frac{28}{9}$

Let's understand Method 2 in detail :-

$\frac{7}{12}, \frac{5}{9}$

We need to have same denominators on both sides.

$\left(\frac{7 \times 3}{12 \times 3}\right), \left(\frac{5 \times 4}{9 \times 4}\right) \rightarrow$ Den. & Num. same Number से Multiply honge

$\frac{21}{36} > \frac{20}{36}$

Then, Numerator ke basis par, smaller & larger decide होगा

Q14. Put in ascending order

$\frac{2}{5}, \frac{3}{7}, \frac{5}{12}$

A) 2/5 < 3/7 < 5/12
B) 3/7 < 5/12 < 2/5
C) 5/12 < 2/5 < 3/7
D) 2/5 < 5/12 < 3/7
E) 5/12 < 3/7 < 2/5
Answer = 2/5 < 5/12 < 3/7

$\frac{2}{5} \quad \frac{3}{7} \quad \frac{5}{12}$

• $\frac{2}{5} \times \frac{3}{7}$ $\left\{\frac{3}{7} > \frac{2}{5}\right\}$

$14 < 15$

• $\frac{3}{7} \times \frac{5}{12}$ $\left\{\frac{3}{7} > \frac{5}{12}\right\}$

$36 > 35$

• $\frac{2}{5} \times \frac{5}{12}$ $\left\{\frac{5}{12} > \frac{2}{5}\right\}$

$25 > 24$

Hence, $\frac{3}{7} > \frac{5}{12} > \frac{2}{5}$

Q15. Put in descending order

$\frac{7}{8}, \frac{11}{12}, \frac{9}{10}$

A) 9/10 > 11/12 > 7/8
B) 11/12 > 9/10 > 7/8
C) 7/8 > 9/10 > 11/12
D) 11/12 > 7/8 > 9/10
E) 7/8 > 11/12 > 9/10
Answer = 11/12 > 9/10 > 7/8

$\frac{7}{8} \quad \frac{11}{12} \quad \frac{9}{10}$

• $\frac{7}{8} \times \frac{11}{12}$ $\left\{\frac{7}{8} < \frac{11}{12}\right\}$

$84 < 88$

• $\frac{11}{12} \times \frac{9}{10}$ $\left\{\frac{11}{12} > \frac{9}{10}\right\}$

$110 > 108$

• $\frac{7}{8} \times \frac{9}{10}$ $\left\{\frac{7}{8} < \frac{9}{10}\right\}$

$70 < 72$

Hence, $\frac{11}{12} > \frac{9}{10} > \frac{7}{8}$

LCM → Least Common Multiple
ऐसा Lowest (Minimum) number जो given numbers के Table पर common हो!

Q16. $\frac{7}{18}+\frac{5}{18}=?$

A) 2/3 B) 11/18 C) 5/6 D) 5/9 E) 13/18

$$\frac{7}{18}+\frac{5}{18}=\frac{7+5}{18}$$

$$=\frac{12}{18}=\frac{6\times2}{6\times3}=\frac{2}{3}$$

Q17. $\frac{13}{25}-\frac{7}{25}=?$

A) 8/25 B) 5/25 C) 6/25 D) 12/25 E) 1/5

$$\frac{13}{25}-\frac{7}{25}=\frac{13-7}{25}$$

$$=\frac{6}{25}$$

Q18. $\left(\frac{11}{12}-\frac{5}{12}\right)-\frac{3}{12}=?$

A) 1/12 B) 1/6 C) 1/3 D) 1/4 E) 5/12

$$\left(\frac{11-5}{12}\right)-\frac{3}{12}$$

$$=\frac{6}{12}-\frac{3}{12}=\frac{6-3}{12}=\frac{3}{12}=\frac{1}{4}$$

Q19. $\left(-\frac{7}{20}\right)+\frac{11}{20}=?$

A) 1/2 B) 2/5 C) 3/10 D) 1/5 E) 1/4

$$\left(\frac{-7}{20}\right)+\frac{11}{20}=\frac{-7+11}{20}$$
$$=\frac{+4}{20}=\frac{1}{5}$$

Q20. $3\frac{4}{9}+1\frac{2}{9}=?$

A) $4\frac{5}{9}$ B) $4\frac{2}{3}$ C) $5\frac{6}{9}$ D) $5\frac{2}{9}$ E) $4\frac{8}{9}$

$$\left(3\frac{4}{9}\right)+\left(1\frac{2}{9}\right)=\left(3+\frac{4}{9}\right)+\left(1+\frac{2}{9}\right)=3+\frac{4}{9}+1+\frac{2}{9}$$
$$=(3+1)+\left(\frac{4+2}{9}\right)=4+\frac{6}{9}=4+\frac{2}{3}=4\frac{2}{3}$$

Q21. $\frac{7}{18}+\frac{5}{12}=?$

A) 29/36 B) 23/36 C) 31/36 D) 17/24 E) 11/18

LCM of 18 and 12

$18\times2=36$
$12\times3=36$

$$\frac{7}{18}+\frac{5}{12}$$

$$\frac{(7\times2)+(5\times3)}{36}=\frac{14+15}{36}=\frac{29}{36}$$

Q22. $\frac{13}{20} - \frac{7}{25} = ?$

A) 49/100 B) 53/100 C) 37/100 D) 17/50 E) 7/20

LCM of 20 and 25

$20 \times 5 = 100$

$25 \times 4 = 100$

$\frac{13}{20} - \frac{7}{25}$

$= \frac{(13\times5)-(7\times4)}{100} = \frac{65-28}{100} = \frac{37}{100}$

Q23. $\left(\frac{5}{6} - \frac{1}{4}\right) - \frac{1}{3} = ?$

A) 1/6 B) 5/12 C) 1/3 D) 1/4 E) 7/12

Answer = 1/4

LCM of 6, 4 and 3

$6 \times 2 = 12$

$4 \times 3 = 12$

$3 \times 4 = 12$

$= \frac{5}{6} - \frac{1}{4} - \frac{1}{3}$

$= \frac{(5\times2)-(1\times3)-(1\times4)}{12}$

$= \frac{10-3-4}{12} = \frac{3}{12} = \frac{1}{4}$

Q24. $2\frac{1}{3} + 1\frac{5}{6} = ?$

A) $3\frac{1}{6}$ B) $4\frac{1}{6}$ C) $4\frac{5}{6}$ D) $3\frac{5}{6}$ E) $5\frac{1}{6}$

$= 2\frac{1}{3} + 1\frac{5}{6}$

$= \left(2+\frac{1}{3}\right) + \left(1+\frac{5}{6}\right) = 2+\frac{1}{3}+1+\frac{5}{6}$

$= (2+1) + \left(\frac{(1\times2)+(5\times1)}{6}\right)$

{ LCM of 3 and 6 will be 6 }

$= 3 + \left(\frac{2+5}{6}\right)$

$= 3 + \frac{7}{6} = 3 + \left(1+\frac{1}{6}\right) = 4\frac{1}{6}$

Q25. $5 - \frac{17}{12} = ?$

$A)\ 3\frac{5}{12}$ $B)\ 3\frac{11}{12}$ $C)\ 4\frac{7}{12}$ $D)\ 2\frac{7}{12}$ $E)\ 3\frac{7}{12}$

$= \frac{5}{1} - \frac{17}{12}$

$= \frac{(5\times12)-(17\times1)}{12} = \frac{60-17}{12} = \frac{43}{12} = 3\frac{7}{12}$

Q26. $\left(2\frac{3}{4}\right) - \left(1\frac{5}{8}\right) = ?$

$A)\ 1\frac{1}{8}$ $B)\ 1\frac{3}{8}$ $C)\ 1\frac{5}{8}$ $D)\ \frac{9}{8}$ $E)\ \frac{7}{8}$

$= \left(2+\frac{3}{4}\right) - \left(1+\frac{5}{8}\right) = 2+\frac{3}{4}-1-\frac{5}{8}$

$= (2-1) + \left(\frac{3}{4}-\frac{5}{8}\right) = 1 + \frac{(3\times2)-(5\times1)}{8} = 1+\frac{1}{8} = 1\frac{1}{8}$

Q27. $\left(\frac{21}{32}\right) \times \left(\frac{16}{63}\right) = ?$

A) 2/7 B) 1/3 C) 2/3 D) 3/8 E) 1/6

$\frac{21}{32} \times \frac{16}{63} = \frac{\cancel{21}^{1}}{\cancel{63}_{3}} \times \frac{\cancel{16}^{1}}{\cancel{32}_{2}} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$

Q28. $\left(\frac{18}{35}\right) \times \left(\frac{21}{24}\right) = ?$

A) 3/5 B) 3/10 C) 7/20 D) 9/40 E) 9/20

$= \frac{18}{35} \times \frac{21}{24}$

$= \frac{9}{20}$

Q29. $\left(\frac{9}{14}\right) \times \left(\frac{28}{45}\right) \times \left(\frac{15}{8}\right) = ?$

A) 4/3 B) 3/4 C) 5/6 D) 2/3 E) 9/8

$\frac{9}{14} \times \frac{28}{45} \times \frac{15}{8} = \frac{9}{45} \times \frac{28}{14} \times \frac{15}{8}$

$= \frac{1}{\cancel{5}_1} \times \cancel{2}^1 \times \frac{\cancel{15}^3}{\cancel{8}_4} = \frac{3}{4}$

Q30. $\left(2\frac{1}{3}\right) \times \left(\frac{3}{7}\right) = ?$

A) 3/2 B) 7/3 C) 3/7 D) 2/7 E) 1

$= \left(2\frac{1}{3}\right) \times \left(\frac{3}{7}\right)$

$= \left[\frac{(2\times3)+1}{3}\right] \times \frac{3}{7}$

$= \frac{7}{3} \times \frac{3}{7} = 1$

Q31. $\left(\frac{5}{8}\right) \div \left(\frac{10}{21}\right) = ?$

A) 21/16 B) 16/21 C) 5/16 D) 2/3E) 3/2

$= \frac{5}{8} \div \frac{10}{21}$

$= \frac{5}{8} \times \frac{21}{10}$ (Reciprocal)

$= \frac{21}{16}$

Q32. $\left(\frac{13}{20}\right) \div \left(\frac{26}{35}\right) = ?$

A) 5/7 B) 7/8 C) 8/7 D) 13/14 E) 14/13

$= \frac{13}{20} \div \frac{26}{35}$

$= \frac{13}{20} \times \frac{35}{26}$

$= \frac{7}{8}$

either X to ÷ or ÷ to X hoga, then value Reciprocal ho jayegi

Q33. $(4\frac{1}{2}) \div \left(1\frac{1}{5}\right) = ?$

A) 15/4 B) 4/15 C) 5/3 D) 3/5 E) 9/4

$= \left(4\frac{1}{2}\right) \div \left(1\frac{1}{5}\right)$

$= \left(\frac{8+1}{2}\right) \div \left(\frac{5+1}{5}\right)$

$= \frac{9}{2} \div \frac{6}{5}$

$= \frac{9}{2} \times \frac{5}{6} = \frac{15}{4}$

Q34. $\left(\frac{5}{6}\right) \div \left(\frac{2}{3}\right) \div \left(\frac{5}{4}\right) = ?$

A) 4/5 B) 2 C) 1/2 D) 5/4 E) 1

$= \frac{5}{6} \div \frac{2}{3} \div \frac{5}{4}$

$= \frac{\cancel{5}}{6} \times \frac{3}{2} \times \frac{4}{\cancel{5}} = \frac{\cancel{3}}{\cancel{6}_2} \times \frac{\cancel{4}^2}{\cancel{2}} = \frac{2}{2}$

$= 1$

Q35. $\left(\frac{2}{5}\right) \div \left(\frac{4}{15}\right) + \frac{1}{3} = ?$

A) 11/9 B) 7/6 C) 9/6 D) 11/6 E) 5/6

$\left(\frac{2}{5} \div \frac{4}{15}\right) + \frac{1}{3}$

{BODMAS rule, pehle Division, then Addition,}

$= \left(\frac{2}{5} \times \frac{15}{4}\right) + \frac{1}{3}$

$= \frac{3}{2} + \frac{1}{3} = \frac{9+2}{6} = \frac{11}{6}$

[LCM of 2 and 3 is 6]

Q36. $\left(\frac{7}{12}\right)$ *of* $132 = ?$

A) 66 B) 72 C) 77 D) 84 E) 88

$= \frac{7}{12} \times 132$

$= \frac{7}{\cancel{12}} \times 11 \times \cancel{12} = 77$

Q37. $\left(\frac{5}{8}\right)$ *of* 144 = ?

A) 80 B) 90 C) 85 D) 95 E) 100

$= \frac{5}{8} \times 144$

$= \frac{5}{\cancel{8}} \times (\cancel{8} \times 18) = 5 \times 18$

$= 90$

Q38. $\left(\frac{3}{5} of\ 250\right) - \left(\frac{2}{3} of\ 150\right) = ?$

A) 40 B) 50 C) 60 D) 70 E) 80

$= \left(\frac{3}{5} \times 250\right) - \left(\frac{2}{3} \times 150\right)$

$= (3 \times 50) - (2 \times 50)$

$= 150 - 100 = 50$

Miscellaneous

Q39. $\left\{\left(\frac{3}{4} + \frac{2}{3}\right)\right\} \div \left\{\left(\frac{5}{6} - \frac{1}{2}\right)\right\} = ?$

A) 17/4 B) 4/17 C) 17/8 D) 7/4 E) 34/7

$= \left\{\frac{3}{4} + \frac{2}{3}\right\} \div \left\{\frac{5}{6} - \frac{1}{2}\right\}$

$= \left\{\frac{9+8}{12}\right\} \div \left\{\frac{5-3}{6}\right\}$

$= \frac{17}{12} \div \frac{2}{6} = \frac{17}{\cancel{12}_{2}} \times \frac{\cancel{6}^{1}}{2} = \frac{17}{4}$

Q40. $\left\{\left(\frac{7}{8}-\frac{3}{10}\right)\right\} \div \left\{\left(\frac{1}{4}+\frac{1}{5}\right)\right\} = ?$

A) 23/9 B) 18/23 C) 23/40 D) 23/18 E) 5/9

$= \left\{\frac{7}{8}-\frac{3}{10}\right\} \div \left\{\frac{1}{4}+\frac{1}{5}\right\}$

$= \left\{\frac{70-24}{80}\right\} \div \left\{\frac{5+4}{20}\right\}$

$= \frac{46}{80} \div \frac{9}{20} = \frac{46}{\cancel{80}_{4}} \times \frac{\cancel{20}^{1}}{9} = \frac{23}{18}$

Q41. $\left\{\left(\frac{5}{12}\right) \div \left(\frac{1}{6}\right)\right\} - \left\{\left(\frac{3}{4}\right) \div \left(\frac{9}{16}\right)\right\} = ?$

A) 11/6 B) 5/6 C) 1/6 D) 7/6 E) 2/3

$= \left\{\frac{5}{12} \div \frac{1}{6}\right\} - \left\{\frac{3}{4} \div \frac{9}{16}\right\}$

$= \left(\frac{5}{\cancel{12}_{2}} \times \frac{\cancel{6}}{1}\right) - \left(\frac{3}{4} \times \frac{16}{9}\right)$

$= \frac{5}{2} - \frac{4}{3} = \frac{15-8}{6} = \frac{7}{6}$

Q42. $\left\{1 \div \left(\frac{1}{2}+\frac{1}{3}\right)\right\} + \left\{1 \div \left(\frac{1}{3}+\frac{1}{6}\right)\right\} = ?$

A) 16/5 B) 8/5 C) 6/5 D) 18/5 E) 10/5

$= \left\{1 \div \left(\frac{3+2}{6}\right)\right\} + \left\{1 \div \left(\frac{2+1}{6}\right)\right\}$

$= \left\{1 \div \frac{5}{6}\right\} + \left\{1 \div \frac{3}{6}\right\}$

$= \left(1 \times \frac{6}{5}\right) + \left(1 \times \frac{6}{3}\right)$

$= \frac{6}{5} + 2 = \frac{6+10}{5} = \frac{16}{5}$

Q43. $\left\{\left(\frac{3}{5}\right) \div \left(\frac{1}{2}\right)\right\} \div \left\{\left(\frac{4}{7}\right) \div \left(\frac{2}{3}\right)\right\} = ?$

A) 5/6 B) 5/7 C) 7/6 D) 6/7 E) 7/5

$= \left\{\frac{3}{5} \div \frac{1}{2}\right\} \div \left\{\frac{4}{7} \div \frac{2}{3}\right\}$

$= \left\{\frac{3}{5} \times \frac{2}{1}\right\} \div \left\{\frac{4}{7} \times \frac{3}{2}\right\}$

$= \left(\frac{6}{5}\right) \div \left(\frac{6}{7}\right) = \frac{6}{5} \times \frac{7}{6} = \frac{7}{5}$

Q44. $\left[\left(\frac{3}{5}\right) \div \left\{1 - \left(\frac{2}{5}\right)\right\}\right] \times \left[\left(\frac{7}{9}\right) \div \left\{1 - \left(\frac{2}{3}\right)\right\}\right] = ?$

A) 7/3 B) 3/7 C) 7/9 D) 9/7 E) 1

$= \left[\frac{3}{5} \div \left(\frac{5-2}{5}\right)\right] \times \left[\frac{7}{9} \div \left(\frac{3-2}{3}\right)\right]$

$= \left[\frac{3}{5} \times \frac{5}{3}\right] \times \left[\frac{7}{9} \times \frac{3}{1}\right]$

$= 1 \times \frac{7}{3} = \frac{7}{3}$

Q45: $3\frac{2}{5} + 4\frac{3}{10} + 2\frac{1}{2} = 6\frac{9}{10} + ? + 1\frac{11}{20}$

A) $1\frac{1}{4}$ B) $1\frac{3}{4}$ C) $2\frac{1}{4}$ D) $2\frac{3}{4}$ E) None of these

$3 + \frac{2}{5} + 4 + \frac{3}{10} + 2 + \frac{1}{2} = 6 + \frac{9}{10} + x + 1 + \frac{11}{20}$

$(3+4+2) + \frac{2}{5} + \frac{3}{10} + \frac{1}{2} = (6+1) + x + \frac{9}{10} + \frac{11}{20}$

$(9-7) + \frac{2}{5} + \frac{3}{10} + \frac{1}{2} - \frac{9}{10} - \frac{11}{20} = x$

$2 + \frac{8 + 6 + 10 - 18 - 11}{20} = x$

$2 + \left(\frac{-5}{20}\right) = x$

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$$2-\frac{1}{4}=x \qquad x = 1+\frac{3}{4}$$

$$1+\left(1-\frac{1}{4}\right)=x \qquad x = 1\frac{3}{4}$$

Q46. $9\frac{1}{6} - 7\frac{5}{8} + 6\frac{1}{3} = \frac{31}{8} + ?$

A) 7/6 B) 6 C) 3/2 D) 4 E) None of these

$$9+\frac{1}{6}-7-\frac{5}{8}+6+\frac{1}{3} = 3+\frac{7}{8}+x \qquad \left\{\frac{31}{8} = 3\frac{7}{8}\right\}$$

$$9+\frac{1}{6}-7-\frac{5}{8}+6+\frac{1}{3}-3-\frac{7}{8} = x$$

$$(9-7+6-3)+\frac{1}{3}+\left(-\frac{5}{8}-\frac{7}{8}\right)+\frac{1}{6}=x$$

$$5+\left(\frac{1}{6}+\frac{1}{3}\right)+\left(-\frac{12}{8}\right)=x$$

$$5+\frac{1}{2}-\frac{3}{2}=x$$

$$5+\left(\frac{1-3}{2}\right)=x$$

$$x = 5-1 = 4$$

Q47. $9\frac{2}{9} + 5\frac{1}{18} - \frac{13}{18} = \frac{488}{?}$

A) 54 B) 18 C) 72 D) 36 E) None of these

$$9+\frac{2}{9}+5+\frac{1}{18}-\frac{13}{18} = \frac{488}{x}$$

$$(9+5)+\frac{2}{9}+\left(\frac{1}{18}-\frac{13}{18}\right)=\frac{488}{x}$$

$$14+\frac{2}{9}-\frac{12}{18}=\frac{488}{x}$$

$$14+\frac{4-12}{18}=\frac{488}{x}$$

$$14-\frac{8}{18}=\frac{488}{x}$$

$$14-\frac{4}{9}=\frac{488}{x}$$

$$\frac{126-4}{9}=\frac{488}{x} \qquad x = \frac{\overset{4}{\cancel{488}}\times 9}{\cancel{122}}$$

$$\frac{122}{9}=\frac{488}{x} \qquad x = 36$$

Q48: $? \times 4\frac{3}{5} = 6\frac{1}{4} \times 3\frac{5}{6} \times 2\frac{2}{3}$

A) $7\frac{1}{5}$ B) $8\frac{1}{3}$ C) $9\frac{2}{5}$ D) $13\frac{8}{9}$ E) None of these

$$x \times \frac{23}{5} = \frac{25}{4} \times \frac{23}{6} \times \frac{8}{3}$$

$$x = \frac{25}{4} \times \frac{\cancel{23}}{6} \times \frac{8}{3} \times \frac{5}{\cancel{23}}$$

$$x = \frac{125}{9}$$

$$x = 13\frac{8}{9}$$

Q49. $5\frac{1}{2} \div 11\frac{2}{5} \div 2\frac{2}{19} = ?$

A) 7/12 B) 11/48 C) 13/24 D) 21/20 E) None of these

$$\left(5\frac{1}{2}\right) \div \left(11\frac{2}{5}\right) \div \left(2\frac{2}{19}\right)$$

$$\frac{11}{2} \div \frac{57}{5} \div \frac{40}{19}$$

$$\frac{11}{2} \times \frac{5}{\cancel{57}_{3}} \times \frac{\cancel{19}}{40}$$ {Reciprocal}

$$= \frac{11}{48}$$

Q50. $1 - \cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{2}}}}} = ?$

(A) 29/96 (B) 9/32 (C) 31/96 (D) 25/96 (E) 29/94

$= 1 - \cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{9/2}}}}$

$\left\{4+\frac{1}{2} = \frac{9}{2}\right\}$

$= 1 - \cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3+\frac{2}{9}}}}$

$\left\{3+\frac{2}{9} = \frac{29}{9}\right\}$

$= 1 - \cfrac{1}{1+\cfrac{1}{2+\frac{9}{29}}}$

$= 1 - \cfrac{1}{1+\cfrac{1}{\frac{67}{29}}}$

$= 1 - \cfrac{1}{1+\frac{29}{67}}$

$= 1 - \cfrac{1}{96/67}$

$= 1 - \frac{67}{96} = \frac{29}{96}$